



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

10-04-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5			
	A.	B	B	D	A	A			
SECTION-I (ii)	Q.	6	7	8	9	10	11	12	13
	A.	A	A,B	A,C,D	B,D	A,C,D	B,D	A,C,D	A,C,D
SECTION-III	Q.	1	2	3	4	5			
	A.	5	7	6	3	6			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5			
	A.	B	B	D	C	C			
SECTION-I (ii)	Q.	6	7	8	9	10	11	12	13
	A.	A,B,D	A,B,C,D	B,C,D	A,B,C,D	A,B,C,D	A,B,C	A,C	A,B,C
SECTION-III	Q.	1	2	3	4	5			
	A.	2	7	8	4	3			

PART-3 : MATHEMATICS

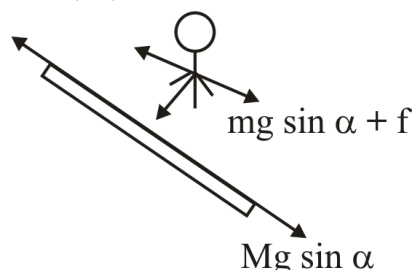
SECTION-I (i)	Q.	1	2	3	4	5			
	A.	B	C	D	C	D			
SECTION-I (ii)	Q.	6	7	8	9	10	11	12	13
	A.	B,C,D	A,C	A,B,C	A,D	A,B,C,D	A,B,C	A,B,C,D	A,B,C,D
SECTION-III	Q.	1	2	3	4	5			
	A.	0	7	9	8	0			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)



F.B.D. of man and plank are

For plank be at rest, applying Newton's second law to plank along the incline

$$Mg \sin \alpha = f$$

$$a = g \sin \alpha \left(1 + \frac{M}{m} \right) \text{ down the incline}$$

Alternate solution :

If the friction force is taken up the incline on man, then application of Newton's second law to man and plank along incline yields.

$$f = Mg \sin \alpha - f = ma \quad 1$$

$$mg \sin \alpha - f = ma \quad 2$$

Solving 1 and 2

$$a = g \sin \alpha \left(1 + \frac{M}{m} \right) \text{ down the incline}$$

Alternate solution :

Application of Newton's second law to system of man + plank

$$a = g \sin \alpha \left(1 + \frac{M}{m} \right) \text{ down the incline}$$

2. **Ans (B)**

$$\frac{2}{1} = \frac{66.9 + \alpha}{33.1 + \beta} \text{ and } \frac{1}{2} = \frac{33.5 + \alpha}{66.5 + \beta}$$

solving we get, $\alpha = -0.1$ cm and $\beta = 0.3$ cm

$$\frac{X}{1} = \frac{40.3 - 0.1}{59.7 + 0.3} = \frac{40.2}{60.0} = 0.67$$

$$t = \frac{9}{5} \text{ s}$$

4. **Ans (A)**

Suppose emf of the battery connected in the circuit is equal to E . Then the current at $R = 0$

will be equal to $I_0 = \frac{E}{R_0}$ when R increases, current through the circuit decreases.

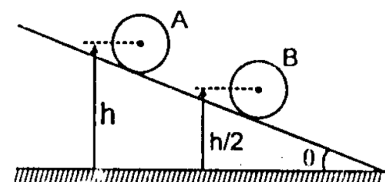
If $R = R_0$, then net resistance of the circuit will become equal to $2R_0$.

Hence, current through the circuit will become

$$\frac{E}{2R_0} = \frac{I_0}{2}$$

To reduce the current to $I_0/4$, the net resistance of the circuit is to be increased to 4 times of its original value R_0 . Hence, $R = 3R_0$. Similarly, to reduce the current to $I_0/8$, the resistance R must be equal to $7R_0$ and to reduce the current to zero, R must be very large. Hence, (A) is correct.

5. **Ans (A)**



Just before collision Between two Balls

potential energy lost by Ball A = kinetic energy gained by Ball A.

$$mg \frac{h}{2} = \frac{1}{2} I_{cm} \omega^2 + \frac{1}{2} m v_{cm}^2$$

$$= \frac{1}{2} \times \frac{2}{5} m R^2 \times \left(\frac{v_{cm}}{R} \right)^2 + \frac{1}{2} m v_{cm}^2$$

$$= \frac{1}{5} m v_{cm}^2 + \frac{1}{2} m v_{cm}^2$$

$$\Rightarrow \frac{5}{7} mgh = m v_{cm}^2 \Rightarrow \frac{mgh}{7} = \frac{1}{5} m v_{cm}^2$$

After collision only translational kinetic energy is transferred to ball B

So just after collision rotational kinetic energy

$$\text{of Ball A} = \frac{1}{5} m v_{cm}^2 = \frac{mgh}{7}$$

PART-1 : PHYSICS

SECTION-I (ii)

6. **Ans (A)**

In the first case, the reading of the scale will increase approximately by the same amount as if the fly was sitting on the bottom of the bottle, i.e. it will increase by the weight of the fly: Since the fly "flutters around" it exerts a force on the air equal to its weight (and the air exerts the same force on the fly), while the air in its turn acts with the same force on the bottom of the bottle.

In the second case the lazy fly + balloon displace an amount (weight) of air which is exactly equal to their weight (that's why it remains airborne) and thus will have no effect on the scale

7. **Ans (A,B)**

For conical pendulum

$$\omega^2 = \frac{g}{\ell \cos \phi} = \frac{g}{L \cos \theta}$$

$$\therefore \cos \phi = \frac{g}{\omega^2 \ell} \quad \dots\dots(1)$$

$$\cos \theta = \frac{g}{\omega^2 L} \quad \dots\dots(2)$$

$$\text{and } \ell \cos \phi = L \cos \theta \quad \dots\dots(3)$$

$$\text{also } T \cos \theta = Mg \text{ and } T \cos \phi = mg$$

$$\therefore \frac{\cos \phi}{\cos \theta} = \frac{m}{M} \quad \dots\dots(4)$$

From (3) and (4)

$$ML = m\ell$$

9. **Ans (B,D)**

As A and C are earthed, they are connected to each other. Hence, 'A+B' and 'B+C' are two capacitors with the same potential difference. If B is closer to A than to C, then the capacitance C_{AB} is $> C_{BC}$. The upper surface of B will have greater charge than the lower surface. As the force of attraction between the plates of a capacitor is proportional to Q^2 , there will be a net upward force on B. This can balance its weight.

10. Ans (A,C,D)

$$19B = \frac{m\theta^2}{r} \text{ or } \frac{g}{m} = \frac{w}{B} = \frac{2\pi}{TB}$$

$$\left(\frac{2\pi}{T}\right) \Delta t = (2n+1)\pi \text{ or } \frac{2\pi}{t} = (2n+1) \frac{\pi}{\Delta t}$$

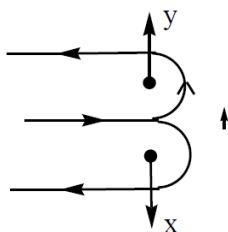
$$\therefore \frac{g}{m} = \frac{(2n+1)\pi}{(\Delta t)B}$$

$$\therefore B = \frac{(2m+1)\pi}{(\Delta t)/(g/m)}$$

$$\vec{B} = \pm \hat{B} B_v$$

$$= \pm \frac{(2n+1)\pi}{(\Delta t)(U/m)}$$

$$= \pm \frac{(2n+1)\pi \hat{B}_v}{\pi \times 10^{-6} \times 2 \times 10^{10}} = \pm (2n+1) 5 \times 10^{-3} \hat{B}_v T$$



11. Ans (B,D)

CONCEPTUAL

12. Ans (A,C,D)

ACD At distance r from left end of the conductor, its horizontal dimension is equal to

$$\left(2x - \frac{x}{\ell}r\right) = \frac{(2\ell - r)}{\ell}x$$

And vertical dimension is equal to

$$\left(y + \frac{y}{\ell}r\right) = \frac{(\ell + r)}{\ell}y$$

$$\Rightarrow \text{area} A = \frac{(2\ell - r)(\ell + r)xy}{\ell^2}$$

If an elemental length dr is considered, its resistance

$$\Rightarrow dR = \frac{\rho dr}{A} = \frac{\rho \ell^2 dr}{(2\ell - r)(\ell + r)xy}$$

$$\Rightarrow R = \int_0^\ell \frac{\rho \ell^2}{xy} = \frac{dr}{(2\ell - r)(\ell + r)} = \frac{2\rho \ell}{xy} \log_e(2)$$

Hence, (A) is correct

Rate of generation of heat per unit length is

$$\text{equal to } i^2 \frac{\rho}{A}.$$

The rate of generation of heat per unit length is maximum where A is minimum.

For the given conductor, cross sectional area A is maximum at its middle section and that is equal to

$$(1.5)x(1.5)y = 2.25xy$$

$\therefore$ rate of generation of heat is minimum

$\therefore$ (B) is wrong

$$\text{Since, } i = n\rho A V_d, \text{ drift velocity } V_d = \frac{i}{n\rho A}$$

$\therefore V_d$ is minimum at middle section

$\therefore$ (C) is correct.

Electric field intensity in current carrying

$$\text{conductor is given by } E = \frac{i\rho}{A}$$

The conductor is made of a uniform material,

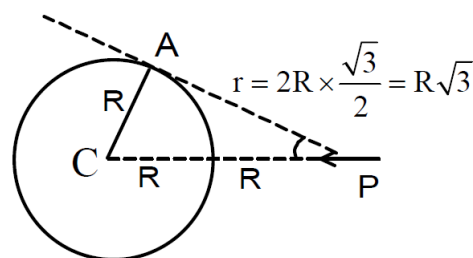
therefore ρ is constant.

A is same at two ends, therefore, electric field

intensity is same at two ends of the conductor.

$\therefore$ (D) is correct.

13. Ans (A,C,D)



Potential at center and at any point on the sphere

will be same

$$\text{So, } V_A = V_C = \frac{P}{4\pi\epsilon_0(2R)^2} = \frac{P}{16\pi\epsilon_0 R^2}$$

$$(V_A)_{\text{dipole}} = \frac{p \cos 30^\circ}{4\pi\epsilon_0 (R\sqrt{3})^2}$$

$$(V_A)_{\text{Dipole}} = \frac{\sqrt{3}P}{24\pi\epsilon_0 R^2}$$

$$(V_A)_{\text{In}} = V_A - (V_A)_{\text{Dipole}}$$

$$(V_A)_{\text{In}} = \frac{P(3 - 2\sqrt{3})}{48\pi\epsilon_0 R^2}$$

PART-1 : PHYSICS

SECTION-III

1. **Ans (5)**

$$x_B = a \sin\left(\frac{2\pi}{6}t\right)$$

$$x_A = a \sin\left(\frac{2\pi}{4}t - \frac{\pi}{2}\right)$$

when they collide

$$a \sin\left(\frac{2\pi}{6}t\right) = a \sin\left(\frac{2\pi}{4}t - \frac{\pi}{2}\right)$$

$$\frac{2\pi}{6}t = \pi \Rightarrow \left[\frac{2\pi}{4}t - \frac{\pi}{2}\right] = \frac{3\pi}{2} - \frac{2\pi}{4}t$$

$$t\left[\frac{1}{2} + \frac{1}{3}\right] = \frac{3}{2} \text{ or } \frac{5t}{6} = \frac{3}{2}$$

2. **Ans (7)**

Signal sent at time t by the source is received by the receiver.

Distance covered by the source in time t ,

$$S = \frac{1}{2}(10)t^2 = 5t^2 \text{ m}$$

$\therefore$ Time elapsed by the sound pulse in reaching

$$\text{to detector} = \frac{5t^2}{340} = \frac{t^2}{68} \text{ sec}$$

$$\Rightarrow t + \frac{t^2}{68} = 10 \Rightarrow t^2 + 68t - 680 = 0 \Rightarrow t = 8.8 \text{ sec}$$

Velocity of the source at t , $v = at = 88 \text{ m/s}$.

$$\text{Thus, } f = 1700 \left(\frac{340}{340 + 88} \right) = 1350 \text{ Hz}$$

3. **Ans (6)**

Applying momentum conservation along the surface of inclined plane.

$$mv_0 \cos 60^\circ + 0 = (m + M)v$$

$$\frac{0.02v_0}{2} = (4.02)v \Rightarrow v = \frac{0.01}{4.02}v_0$$

Where v is the velocity of bullet block system just after the collision.

Acceleration of block bullet + system along the incline

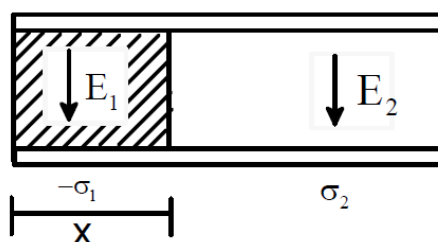
$$a = -g \sin 30^\circ = -5 \text{ m/s}^2$$

By $v = u + at$

$$0 = \frac{0.01}{4.02}v_0 - 5(12) \Rightarrow v_0 = 24.120 \text{ m/s}$$

4. **Ans (3)**

(a)



$E_1 = E_2$, since the potential difference between the plates is same.

$$\Rightarrow \frac{\sigma_1}{\epsilon_0 \epsilon_0} = \frac{\sigma_2}{\epsilon_0} \Rightarrow \sigma_1 = 4\sigma_2$$

$$\text{Also } 4\sigma_2 \ell x + \sigma_2 \ell (\ell - x) = Q$$

$$\Rightarrow \sigma_2 = \frac{Q}{\ell^2 + 3\ell x} \quad \sigma_1 = \frac{4Q}{\ell^2 + 3\ell x}$$

$$\Rightarrow \sigma_1 = \frac{4Q}{\ell^2 + 3\ell \times \frac{\ell}{3}} = \frac{2Q}{\ell^2}$$

$$Q_1 = \sigma_1 \ell x = \frac{\sigma_1 \ell^2}{3}$$

$$Q_1 = \frac{2Q}{3} = 4\mu\text{C}$$

$$\text{bound charge} = 4(1 - 1/k) = 3$$

5. **Ans (6)**

$$\left(P_0 + \frac{4S}{R}\right) \left(\frac{4}{3}\pi R^3\right) = \left(\frac{P_0}{2} + \frac{4S \times 4}{5R}\right) \times \frac{4}{3}\pi \left(\frac{5R}{4}\right)^3$$

$$\Rightarrow P_0 + \frac{4S}{R} = \frac{125P_0}{128} + \frac{25S}{4R}$$

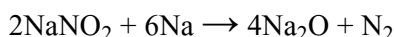
$$\Rightarrow P_0 = \frac{96S}{R} = \frac{16nS}{R}$$

$$\therefore n = 6$$

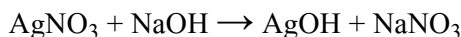
PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (B)**



2. **Ans (B)**



Brown

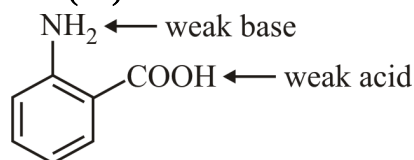
3. **Ans (D)**

Cumene hydroperoxide reaction

4. **Ans (C)**

Decarboxylation of β -keto acid

5. **Ans (C)**



weak acid & weak base $\rightarrow$ no reaction

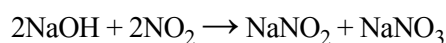
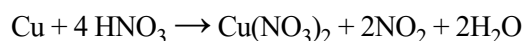
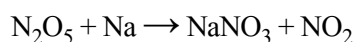
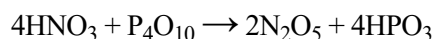
PART-2 : CHEMISTRY

SECTION-I (ii)

6. **Ans (A,B,D)**



7. **Ans (A,B,C,D)**



Liquid N_2O_5 autoionises as $N_2O_5 \rightarrow NO_2^+ + NO_3^-$

The substance that increase anion (NO_3^-) in a solvent is base.

The substance that increase cation (NO_2^+) in a solvent is acid.

8. **Ans (B,C,D)**

After removing an electron, the remaining ns electron goes into $(n-1)d$ orbital exception Mn^+ as it has d^5 configuration Ref. NCERT Page No. 215

9. **Ans (A,B,C,D)**

A) Due to small size and more effective nuclear charge. The transition metal ions have more polarizing power than s- block metal ions having comparable size and same oxidation state

B) As the size of halide ion increases polarisability increases. So covalent character of silver halides increases with increase in size of halide ion. Thus their solubility in water decreases

C) LiCl is covalent soluble in organic solvents but NaCl is ionic insoluble in organic solvents

D) Oxo salts means salts of oxo acids in case of polyatomic anion as covalent nature increases thermal stability decreases.

10. **Ans (A,B,C,D)**

More stable carbocation implies more difference b/w its transition state and its own energy level also CN^{1-} in large excess will generate a pseudo uni molecular reaction making difficult to do experimental determination

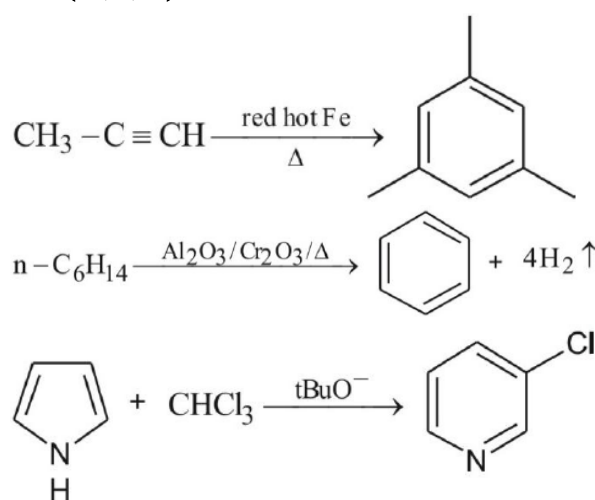
11. **Ans (A,B,C)**

Steric crowd of tert butyl group do not support ortho % to be high

12. **Ans (A,C)**

Ethers are not reduced by Raney Ni.

13. **Ans (A,B,C)**



PART-2 : CHEMISTRY

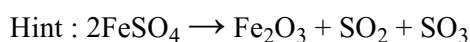
SECTION-III

1. **Ans (2)**

Sum of oxidation numbers of sulphur in hypo is $-2 + 6 = +4$

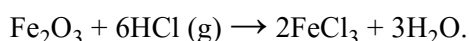
Oxidation number of sulphur in the product $NaHSO_4(X)$ formed by the oxidation of hypo with Cl_2 is +6. Difference is 2

2. **Ans (7)**



Gas B is SO_2 since it reduces acidified $K_2Cr_2O_7$

Gas C is SO_3 . Its trimer contain 3 S – O – S bonds (m)



$FeCl_3$ exists dimer in vapour phase. So it

contain 4 M – Cl bonds (n). Total (m + n) is 7

3. **Ans (8)**

cationic part having two optical isomers.
Anionic part having four optical isomers so collectively compound having total 8 optical isomers.

4. **Ans (4)**

Conceptual

5. **Ans (3)**

1, 3, 4 are correct statements

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B)**

Product of the perpendicular distances,
calculated from foci, upon any tangent is b^2

$$\Rightarrow 3d = 7 \times 7 \Rightarrow \sqrt{3d} = 7$$

2. **Ans (C)**

$$\frac{f'(x)(1+x^2) - f(x)2x}{(1+x^2)^2} = \frac{2f(x)}{1+x^2}$$

$$\Rightarrow \frac{f'(x)}{(1+x^2)} = 2 + \frac{2x}{1+x^2}$$

$$\Rightarrow \ln f(x) = 2x + \ln(1+x^2) + c$$

$$\text{as } f(0) = 2 \Rightarrow \ln 2 = c$$

$$\Rightarrow f(x) = 2(1+x^2)e^{2x}$$

3. **Ans (D)**

$$\frac{1}{\sqrt{t} + \sqrt{t^2 - 1}} = \sqrt{\frac{t+1}{2}} - \sqrt{\frac{t-1}{2}}$$

4. **Ans (C)**

After drawing graph of $g(x) = \max |y^2 - xy|$,

$y \in [0, 1]$, we get minimum of $g(x)$ is

$$3 - 2\sqrt{2} \text{ at } x = 2\sqrt{2} - 2.$$

5. **Ans (D)**

$$\binom{n}{0} \binom{n}{1} \binom{n}{2} \dots \binom{n}{n} = \prod_{k=1}^n \left(\frac{n+1-k}{n+1} \right)^{n+1-2k}$$

$$\Rightarrow \left(\binom{n}{0} \binom{n}{1} \binom{n}{2} \dots \binom{n}{n} \right)^{\frac{1}{n(n+1)}} = \left(\prod_{k=1}^n \left(1 - \frac{k}{n+1} \right)^{n+1-2k} \right)^{\frac{1}{n(n+1)}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n} \ln \left(\binom{n}{0} \binom{n}{1} \binom{n}{2} \dots \binom{n}{n} \right)^{\frac{1}{n(n+1)}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(1 - \frac{2k}{n+1} \right) \ln \left(1 - \frac{k}{n+1} \right)$$

$$= \int_0^1 (1-2x) \ln(1-x) dx = \frac{1}{2}$$

Hence required limit $\sqrt{e}$

PART-3 : MATHEMATICS

SECTION-I (ii)

6. **Ans (B,C,D)**

$$z = x + iy$$

$$\frac{2(x+iy) - (x-iy)}{3} = a - \frac{1}{3} + ib.$$

$$\& a^2 - b^2 = 1$$

$$\Rightarrow \frac{(x+1)^2}{9} - y^2 = 1$$

$$\& e = \frac{\sqrt{10}}{3}.$$

7. **Ans (A,C)**

$$p + q = 2\sqrt{pq} = 10$$

$$\text{Also } p^2 + q^2 \geq 2pq = 50$$

8. **Ans (A,B,C)**

Let P be $(3\cos\theta, \sin\theta + 4)$.

$A(x_1, x_1)$ & $B(x_2, x_2)$ & x_1, x_2 are the roots of x^3

$$+ 4x - 4 = 0$$

Area of ABP, is

$$= \frac{1}{2} |3\cos\theta - \sin\theta - 4| |x_1 - x_2|$$

$$\Delta_1 = 2\sqrt{2} (4 - \sqrt{10})$$

$$\Delta_2 = 2\sqrt{2} (4 + \sqrt{10})$$

9. **Ans (A,D)**

use $\sin^3 x = \frac{3}{4} \sin x - \frac{1}{4} \sin 3x$ to get

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n 3^{r-1} \sin^3 \frac{x}{3^r} = \frac{1}{4} (x - \sin x)$$

12. **Ans (A,B,C,D)**

By applying LMVT we get all A,B,C,D correct.

13. **Ans (A,B,C,D)**

We can easily see that C_1 and C_2 are inverse function of each other. Hence their graphs will be mirror images of each other in a line mirror $y = x$. Draw the graph of $C_1 : y = a^x$ and

$C_2 : x = a^y ; a > 0$ and $y = x$. After analyzing the graph we can observe the following things; For $a \in (0, e^{-e})$, number of times C_1 meets with C_2 is 3. For $a \in [e^{-e}, 1] \cup \{e^{e^{-1}}\}$, number of times C_1 meets with C_2 in

1. For $a \in \{1, e^{e^{-1}}\}$, number of times C_1 meets with C_2 is 2. For $a \in \{e^{e^{-1}}, \infty\}$, number of times C_1 meets with C_2 is 0.

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (0)**

Substitute $x = \frac{ab}{t}$.

2. **Ans (7)**

Using tr. $A = 50$ we get $\det A = 46 \Rightarrow \lfloor \sqrt{k} \rfloor = 7$

3. **Ans (9)**

$$A^n = \begin{bmatrix} 2^n & n2^{n-1} & \frac{n(n-1)}{2} 2^{n-2} \\ 0 & 2^n & n2^{n-1} \\ 0 & 0 & 2^n \end{bmatrix}$$

$$\Rightarrow A^{10} = \begin{pmatrix} 1024 & 5120 & 11520 \\ 0 & 1024 & 5120 \\ 0 & 0 & 1024 \end{pmatrix}$$

$$\left\lfloor \frac{k}{2560} \right\rfloor = \left\lfloor \frac{3 \times 4 + 2 \times 20 + 45}{10} \right\rfloor$$

$$= \left\lfloor \frac{12 + 40 + 45}{10} \right\rfloor = 9$$